INCH-POUND

A-A-3085A
February 5, 2002
SUPERSEDING
A-A-3085
November 7, 1996

COMMERCIAL ITEM DESCRIPTION

PROTRACTORS, MECHANICS, UNIVERSAL BEVEL AND VERNIER SCALE

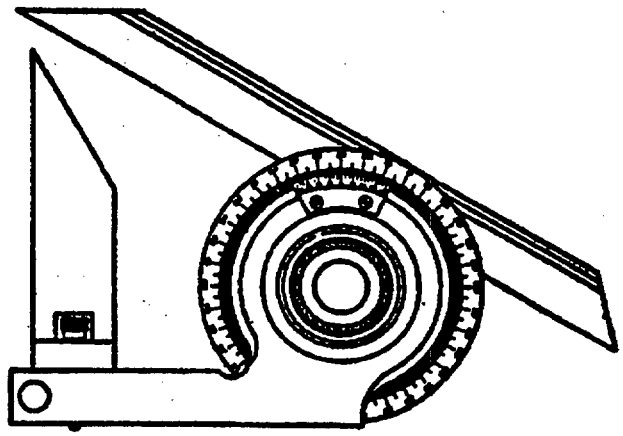
The General Services Administration has authorized the use of this Commercial Item Description by all federal agencies.

1. SCOPE. This Commercial Item Description covers protractors designed for precision measuring, checking, laying out of angular work, and scribing layout lines.

2. CLASSIFICATION. Not applicable.

3. SALIENT CHARACTERISTICS.

3.1 Design. The universal bevel, vernier scale protractor shall have a frame, graduated dial, thumbnut, and adjustable blade with an acute angle attachment and revolving dial. The protractor shall be made of stainless or corrosion resistant steel, heat treated to produce a protractor capable of long life under service conditions.



Protractor, universal bevel, acute angle attachment, revolving dial.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Hardware SuperStore, 6FEE, 1500 Bannister Rd., Bldg# 6, Kansas City, Missouri. 64131.

FSC 5210

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3.1.1 Accuracy. The following tests for accuracy shall be made with suitable equipment of proven accuracy, as certified by the National Institute of Standards and Technology (NIST). All tests shall be performed at a constant temperature of 68 degrees F, +/- 10 degrees F.

3.1.2 Vernier scale. The vernier scale shall be made to fit neatly on the beveled edge of the frame and shall be held rigidly in position. The graduations shall extend to the outer surface edge of the scale and shall be made and fitted so as to minimize parallax error in reading between the vernier scale and the main scale. There shall be a clearance of not more than .006 inch between the bottom surface of the vernier scale and the surface of the main scale directly beneath the vernier scales' outer lower edge. The vernier scale shall be graduated in 5-minute increments from 0 to 60 minutes both left and right from the 0 index. The graduations shall be cut clearly and straight to a width of not less than .003 inch or more than .006 inch, and to a depth less than .001 inch or more than .003 inch. The 0 minute and each succeeding 15-minute shall be marked with the corresponding number and shall be longer than the intermediate graduations. The intermediate graduations shall be readable to 5-minute increments. The vernier scale shall have an accuracy of +/- 1 minute between any two graduations. In addition, each graduation marking shall be within 5 minutes of its true reading and verified by means of either a coordinate measuring machine, a sine bar, or angular blocks to ascertain the total angular error.

3.1.3 Acute-angle attachment. The acute angle attachment shall fit readily to the base, and shall be capable of being clamped at any desired point within the limit provided in the base. When clamped the longitudinal axis shall be perpendicular within +/- 2 minutes to the base in the same plane as the blade and verified with a test indicator in conjunction with a height gage and surface plate.

3.1.4 Frame. The frame of the protractor shall be of one piece construction and so constructed that it will lie flat upon the work. The squareness shall be determined by means of a precision square and gage blocks. The protractor shall essentially consist of a circular member upon which is mounted to the vernier scale with a base whose top and bottom are at right angles to the diameter of the circular section passing through zero graduation on the vernier scale. At zero setting, the measuring edges of the blades and the base shall be parallel within +/- 2 1/2 minutes. The measuring edges shall be flat within .0005 inch over the length of the frame and verified with a test indicator in conjunction with a height gage and surface plate. The frame shall not have a value less than 92 on the Rockwell B scale when tested per ASTM-E18.

3.1.5 Dial. The dial shall not be less than 2 ½ inches or more than 3 inches in diameter and shall have a scale which is accurately graduated in degrees on the outer periphery of the front surface, and shall rotate in a complete circle. The scale shall be divided into four quadrants, each having 90 graduations. The 0 and each 10-degree graduation in either direction shall be marked with the corresponding number. The 5-degree graduations shall be longer than the intermediate graduations. A tolerance of 2 ½ minutes, non-cumulative between any two graduations shall be verified by means of either a coordinate measuring machine, a sine bar, or angular blocks to ascertain the total angular error allowed. The width and depth of the graduations shall be cut clearly and straight to a width of not less than .003 inch or more than .006 inch, and to a depth less than .001 inch or more than .003 inch.

3.1.6 Thumbnuts. Thumbnuts or other devices shall be furnished for locking the rotating dial to the frame and locking the blade at any point.

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3.1.7 Setting. A positive means shall be furnished for securing fine adjustments and smooth movement of the dial in either direction.

3.1.8 Adjustable blades. Adjustable blades shall be capable of being moved smoothly back and forth through their entire length and shall be interchangeable. The blades shall have a longitudinal groove parallel with the edges to facilitate the clamping to the revolving dial and shall have a hardness value not less than 92 on the Rockwell B scale when tested per ASTM-E18. The ends of the blades shall be machined to angles with one edge to 45 degrees and the other end to 60 degrees. Blades of 6 1/2 and 12 inches +/- ½ inch shall be furnished. The blades shall be flat and parallel within .0005 inch and with a test indicator in conjunction with a height gage and surface plate.

3.1.9 Instruction leaflet. Each protractor shall include a clear and concise operational instruction leaflet.

3.2 Identification marking. The tool shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture, unless otherwise specified. All identification markings shall be etched, engraved, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

3.3 Workmanship. Details of workmanship shall be in accordance with the best commercial practices. Paints, coatings, plating, and finishes shall be smooth, adherent, continuous and not stained or discolored. External surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be free from manufacturing workmanship defects (e.g. sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g. pits, rips, fins, burrs, tears, nodules, cracks, or blisters), which may adversely impact the item's serviceability, durability, safety, or appearance

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The supplier/contractor is encouraged to use materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Metric products. Products to metric dimensions will be considered on a equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified (using conversion tables contained in the latest revision of Federal Standard 376) and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units; a request should be made to the contracting officer to determine if the product is acceptable.

5. PRODUCT CONFORMANCE.

5.1 Product conformance. The product provided to the Government shall conform to the contractually specified required. For product characteristics and requirements the provided product shall conform to the suppliers own drawings, standards, specifications, and quality assurance practices for the highest quality product commercially offered by the supplier. If the

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supplier does not commercially offer the product, the best commercial practices shall apply. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as provided for under the provisions of the contract.

5.2 Responsibility for inspecting. Unless otherwise specified, the supplier is responsible for the performance of all inspection requirements and may use any commercial facilities (including the suppliers own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any inspection deemed necessary to assure the item conforms to the specified requirements.

6. PACKAGING. Requirements of preservation, packing, packaging, and marking of the packages shall be specified in the contract or order.

7.NOTES. This section contains information of a general or explanatory nature that is not mandatory.

7.1 Addresses for obtaining copies of referenced documents.

Federal Acquisition Regulation (FAR): Government Printing Office, Superintendent of Documents, Washington, DC 20401-9371

GSA Specification Section (3FP-E), Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC, 20407.

American Society for Testing and Materials (ASTM), 100 Barr Harbor Drive, West Conshohocken, PA, 19428-2959.

National Institute of Standards and Technology (NIST), Public Inquiries Unit, 100 Bureau Drive, Washington DC 20899-2100

7.2 National Stock Numbers (NSN). The following NSN corresponds to the item specified in this CID. This NSN may not be indicative of all possible NSNs associated with this CID.

NSN: 5210-00-390-5610

7.3 Ordering data. Purchasers should select the preferred options permitted herein and should include the following information in procurement documents:

(a) Title, number, and date of Commercial Item Description.
(b) When applicable, the appropriate NSN.

7.4 Key words. Universal Bevel and Vernier Mechanics Protractor.

MILITARY INTERESTS:

NONE: DoD has no registered interest of revisions or notices to this Commercial Item Description until further notice.

Preparing Activity:

GSA - FSS

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